

Neural Networks – Hands-on Assignment

(Regression & Classification)

Objective

Build **end-to-end Neural Network models** for **regression and classification**, focusing on **conceptual understanding** and correct workflow rather than just code.

Dataset Requirements

Create **two synthetic, realistic datasets**:

1 Regression Dataset

- Numerical features
- At least one categorical feature
- Continuous target variable
- Missing values (intentional)
- Few outliers
- Features on different scales

 Example: House price, insurance cost, customer spend.

2 Classification Dataset

- Numerical features
- At least one categorical feature
- Categorical target (binary or multi-class)
- Class imbalance (intentional)
- Missing values and outliers
- Features on different scales

 Example: Churn prediction, loan default, disease risk.

Steps to Follow

1. Data Creation

- Generate realistic synthetic data
 - Briefly explain features, target, and problem relevance
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2. Exploratory Data Analysis (EDA)

- Visualize distributions and relationships
 - Identify missing values, outliers, scaling issues
 - Check class imbalance (classification)
 - Explain why **feature scaling is critical** for Neural Networks
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3. Data Preprocessing

- Handle missing values (justify choice)
 - Encode categorical variables
 - Treat or justify outlier handling
 - Scale features
 - Split into train, validation, and test sets
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4. Neural Network – Regression

- Build a feed-forward NN
 - Choose appropriate:
 - Architecture
 - Activation functions
 - Loss & optimizer
 - Train and track training vs validation loss
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5. Neural Network – Classification

- Build a NN classifier
 - Use correct output activation & loss function
 - Train and monitor loss and accuracy
 - Handle class imbalance if needed
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6. Model Evaluation

Regression:

- MAE, RMSE/MSE, R^2

- Predicted vs actual comparison

Classification:

- Confusion Matrix
- Precision, Recall, F1-score
- ROC-AUC (if binary)

📌 Explain why evaluation differs between regression and classification.

7. Key Observations

- What worked and what didn't
 - Impact of:
 - Network depth
 - Learning rate
 - Epochs
 - Identify underfitting vs overfitting
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★ Try this also additionally...

- Try Dropout or Early Stopping
 - Compare NN performance with Linear / Logistic Regression
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✅ Expectation

- Emphasis on **understanding & explanation**
- Clear reasoning for every decision
- Code **plus** observations matter